ModelIt!

Teacher's Guide

Can We Colonize Mars?

Modeling Life Support Systems and Planetary Engineering Challenges

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-ESS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-4

Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-component computational model to simulate the interdependent life support systems required for a Mars colony and test system resilience under crisis scenarios.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Students analyze how variations in solar energy input, atmospheric composition, and surface conditions on Mars create climate challenges fundamentally different from Earth that constrain all engineering solutions.
Crosscutting Concept	**Systems and System Models** — Students identify how the behavior of the Mars colony emerges from the interactions among ten components rather than from any individual component, revealing that system-level properties like viability cannot be predicted from parts alone.

CAST Testing Alignment

MULTIPLE CHOICE:

During a three-month Martian dust storm, solar energy input drops by 90%. Based on the colony life support model, which system fails first and why does this trigger a cascade through other systems?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a Mars colony that achieves 95% water recycling efficiency is likely unsustainable long-term, while a colony at 98.5% efficiency might survive. Reference at least three interacting components and their feedback relationships.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Mars is the most Earth-like planet in our solar system, yet it remains profoundly hostile to human life. With an atmosphere less than 1% as dense as Earth's, composed of 95% carbon dioxide, surface temperatures averaging negative 60 degrees Celsius, and no global magnetic field to deflect lethal cosmic radiation, every breath a human takes on Mars must be engineered. The question of Mars colonization is ultimately a question of systems engineering at a scale and complexity never before attempted.

The Martian Environment

Mars presents a suite of challenges that interact in complex ways. Surface atmospheric pressure is about 600 pascals — less than 1% of Earth's sea level pressure and below the Armstrong limit where water boils at body temperature. The atmosphere is 95.3% CO2, 2.7% nitrogen, and 1.6% argon with only trace oxygen. Surface gravity is 38% of Earth's, which may cause bone density loss, muscle atrophy, and cardiovascular deconditioning over time. The average surface temperature is negative 60 degrees Celsius with extremes from negative 140 to positive 20. Global dust storms can envelope the entire planet for months, reducing solar energy to near-zero levels. Without a global magnetic field, the surface receives approximately 0.67 millisieverts of radiation per day from galactic cosmic rays and solar particle events.

Life Support Engineering

A Mars colony life support system must replicate the essential functions that Earth provides for free: breathable atmosphere, drinkable water, food, temperature regulation, radiation protection, and waste processing. The International Space Station demonstrates that humans can survive in space for over a year, but ISS depends on regular resupply from Earth — approximately 6,000 kilograms per crew rotation. A Mars colony cannot rely on resupply because launch windows occur only every 26 months and transit takes 6-9 months. The system must therefore be closed-loop: recycling water at 98%+ efficiency, regenerating oxygen from CO2, growing food hydroponically, processing all waste into reusable materials, and maintaining atmospheric pressure and composition within narrow human tolerance ranges.

In-Situ Resource Utilization

The economics and physics of launching materials from Earth to Mars make ISRU essential. Every kilogram launched to Mars costs approximately $100,000-$500,000 depending on the mission architecture. NASA's MOXIE experiment on the Perseverance rover successfully demonstrated CO2-to-oxygen conversion on the Martian surface in 2021, producing about 12 grams of oxygen per hour. Scaling this to colony needs requires approximately 1,000 times MOXIE's capacity. Water extraction from subsurface ice deposits at the mid-latitudes or polar regions could provide the hydrogen needed for rocket fuel (via the Sabatier reaction combining H2 and CO2 to produce methane and water) and the feedstock for electrolytic oxygen production. Regolith processing for construction materials, radiation shielding mass, and potentially even metals is being actively researched.

The Redundancy Imperative

Perhaps the most critical lesson from aerospace engineering is that any system without redundancy will eventually fail. On Earth, system failures are inconveniences; on Mars, they are potentially fatal. Every critical life support function must have at least two independent pathways. Oxygen can be produced by CO2 electrolysis and by water electrolysis. Water can be sourced from recycling and from ice extraction. Energy can come from solar panels and nuclear reactors. Food can be grown in hydroponics and supplemented with stored reserves. The engineering challenge is not making any single system work — it is making ten interdependent systems work simultaneously, continuously, for decades, with sufficient redundancy to survive the failure of any individual component.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the Mars colony life support system: solar energy input, atmospheric CO2, subsurface ice, radiation exposure, oxygen production, water recycling, food production, habitat integrity, waste processing, and colony viability.

E **Establish Relationships**

Students map the causal relationships showing how solar energy drives oxygen, food, and water systems, how radiation creates hard survival thresholds, and how water recycling efficiency compounds across all subsystems.

V **Visualize & Model**

Students build a ten-component computational model showing the interdependent life support architecture with cascading dependencies, feedback loops, and threshold effects.

E **Evaluate Outcomes**

Students run optimal conditions, dust storm crisis, and aging colony scenarios to identify which variables most strongly determine colony viability and where critical failure thresholds exist.

R **Revise & Extend**

Students add psychological stress, regolith toxicity, or communication delay to model how human factors and environmental hazards compound the engineering challenges of Mars colonization.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2020, NASA landed a helicopter on Mars. SpaceX is building a rocket to send humans there by the 2030s. But landing is the easy part. The real question is: can we keep 100 people alive on a planet that is trying to kill them every second of every day?"*

Watch For

Show cover image. Ask students to name everything they know about Mars conditions. Build a class list of challenges on the board.

Slide 2: Learning Objectives

3 min

Say: *"Today you are engineering the most complex life support system ever conceived. You will model ten interacting components and discover why Mars colonization is the ultimate systems engineering challenge."*

Watch For

Have students read objectives. Pre-teach 'closed-loop life support' and 'in-situ resource utilization' with ISS examples. Emphasize the scale difference between ISS (6 people) and a colony (100 people).

Slide 3: Big Question

3 min

Say: *"Mars has no breathable air, no liquid water, lethal radiation, and temperatures that would freeze you solid. Is building a permanent colony there science or science fiction? By the end of today, your model will give you a data-driven answer."*

Watch For

Quick-write: Students list the top three challenges they think are hardest to solve and why. Share out and rank by class consensus.

Slide 4: LEVER Framework

2 min

Say: *"We will Label the ten components of a Mars life support system, Establish their relationships, Visualize the model, Evaluate it under crisis scenarios, and Refine it by adding human factors."*

Watch For

Preview each LEVER step. Emphasize that this is the most complex model in the curriculum — ten components with cascading dependencies.

Slide 5: Activity 1: Components

8 min

Say: *"These are the ten variables that determine whether a Mars colony lives or dies. Four come from Mars itself — the inputs we cannot control. Six are engineered systems we must build and maintain. Sort them."*

Watch For

Students sort external versus internal components. Use think-pair-share to debate borderline cases. Discuss why radiation is external but shielding is part of habitat integrity.

Slide 6: Activity 2: Connections

8 min

Say: *"Solar energy drives oxygen production, food growth, water recycling, and waste processing. But what happens when a dust storm cuts solar by 90%? Trace the cascade."*

Watch For

Students draw arrows showing positive and negative relationships. Identify the cascade chain from energy loss. Discuss which feedback loops could stabilize or destabilize the system.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now test your model under three scenarios: optimal conditions, a global dust storm, and an aging colony after ten years. Record predictions before you run each scenario."*

Watch For

Students run all three scenarios, recording predictions versus observations. Emphasize the dust storm cascade. Discuss: at what point does the Colony Viability Index cross the survival threshold?

Slide 8: Discoveries

5 min

Say: *"Your model reveals three critical truths: energy is the master variable, water recycling efficiency compounds across all systems, and redundancy is not optional — it is survival."*

Watch For

Class discussion of discoveries. Compare initial challenge rankings to model evidence. Ask: which finding surprised you most?

Slide 9: STEM Challenge

12 min

Say: *"Now your team must design a life support system for 100 people that can survive a six-month dust storm with no resupply. Your model data tells you the requirements. Your engineering creativity must meet them."*

Watch For

Groups design integrated life support architectures. Each team must specify energy, water, oxygen, food, radiation, and waste solutions with redundancy. Present and evaluate designs against model-derived requirements.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Solar Energy Input, Atmospheric CO2 Availability, Subsurface Ice Reserves, Radiation Exposure Level
- **INTERNAL:** Oxygen Production Rate, Water Recycling Efficiency, Food Production Capacity, Habitat Structural Integrity, Waste Processing Rate, Colony Viability Index

Relationships

- **Solar Energy Input to Oxygen Production Rate:** POSITIVE (+) — Solar energy powers the electrolysis systems that split CO2 or water to produce oxygen. When solar input drops during dust storms, oxygen production decreases proportionally unless backup power is available.
- **Solar Energy Input to Food Production Capacity:** POSITIVE (+) — Hydroponic and aeroponic growing systems require artificial lighting powered by solar energy. Reduced solar input means less light for photosynthesis and less power for climate control in growing modules.
- **Atmospheric CO2 Availability to Oxygen Production Rate:** POSITIVE (+) — Carbon dioxide is the feedstock for MOXIE-type electrolysis systems. Higher atmospheric CO2 availability provides more raw material for oxygen production, though the thin atmosphere limits collection rates.
- **Subsurface Ice Reserves to Water Recycling Efficiency:** POSITIVE (+) — Access to ice reserves provides a buffer that reduces the pressure on recycling systems. When ice is available, the system can tolerate lower recycling efficiency; when reserves are depleted, recycling must approach 100%.
- **Radiation Exposure Level to Habitat Structural Integrity:** NEGATIVE (-) — Prolonged radiation exposure degrades certain habitat materials and forces colonists to spend more time indoors, increasing wear on internal systems. Extreme radiation events may require emergency shelter protocols that stress structural systems.
- **Water Recycling Efficiency to Food Production Capacity:** POSITIVE (+) — Food production requires irrigation water. Higher recycling efficiency means more water available for hydroponics. A 5% drop in recycling efficiency can reduce food production by 15-20% as water is rationed to drinking and oxygen generation first.
- **Oxygen Production Rate to Colony Viability Index:** POSITIVE (+) — Breathable atmosphere is a non-negotiable requirement. If oxygen production falls below the minimum threshold for human respiration, colony viability drops to zero regardless of all other system performance.
- **Waste Processing Rate to Water Recycling Efficiency:** POSITIVE (+) — Waste processing recovers water from biological waste, graywater, and humidity. Faster waste processing feeds more water back into the recycling system, improving overall water availability for the colony.

Key Discoveries

- Energy is the master variable — when solar input drops during dust storms, every other life support system degrades in a predictable cascade: oxygen production falls first, then food production, then water recycling loses power for purification pumps
- Water recycling efficiency is the most critical internal variable because water is used in oxygen generation, food production, cooling systems, and hygiene — even a 5% drop in recycling efficiency creates compounding resource deficits
- Radiation shielding is a binary threshold — below adequate shielding, cumulative radiation damage reduces crew capacity and eventually becomes lethal, creating a hard constraint that cannot be compensated by other systems
- True self-sufficiency requires redundancy in every system — the model reveals that a colony dependent on a single oxygen source, single food system, or single energy source will eventually fail when that system degrades

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can just terraform Mars to make it Earth-like"	Terraforming Mars would require centuries to millennia of sustained effort. Mars has lost most of its atmosphere to solar wind stripping because it lacks a global magnetic field. Even if we could release all CO_2 locked in polar caps and regolith, atmospheric pressure would reach only about 7% of Earth's — far below breathable levels. Creating a breathable oxygen atmosphere would require converting megatons of CO_2 over thousands of years.
"The main challenge is getting to Mars"	Landing humans on Mars is a solved engineering problem in principle — the challenge is keeping them alive after they arrive. The 6-9 month transit is difficult but finite. Living on Mars indefinitely requires solving closed-loop life support, radiation protection, food production, medical care without hospitals, and psychological endurance — all simultaneously and without resupply for years.
"Mars is basically like a cold desert on Earth"	Mars is far more hostile than any desert on Earth. The atmosphere is unbreathable. The pressure is so low that exposed liquid water boils instantly. The soil contains toxic perchlorates. Radiation levels are 200 times higher than Earth's surface. There is no emergency rescue possible. The psychological analogy to a desert minimizes the engineering challenges by many orders of magnitude.